

Infant Elfrick Gnanasusairaj

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EDUCATION

University of Wisconsin-Madison (4.0/4.0)

Bachelor of Aerospace Engineering + Computer Science

May 2028

Madison, WI

EXPERIENCE

SBEL Student Researcher

Apr 2026 – Present

UW - Madison

Madison, WI

- Independently designed hardware/software for a draw-bar pull testing rig in order to address a gap in collected data. developed closed-loop feedback controllers and state machines for consistent pulling force, while creating a robust data stream to make data integrable with other sensor suites.
- Collaborated with researchers to validate physics-based simulations through experimental testing. Using optimization tools like FEA and custom algorithms to reduce cost, designed a custom inclining sandbox to compare simulated and real-world behavior.
- Designed and implemented hardware/software systems for scale models of machines, including lunar landers, to generate experimental data and improve simulation parity with real testing.
- Presented process, technology, and results at the Machine-Ground Interaction Consortium(MaGIC), a conference focusing on the intersection of simulation, embodied AI, high-performance computing, etc. Involved with development of research papers

Design Innovation Lab AED Drone Dev

Nov 2025 – Present

UW - Madison

Madison, WI

- Worked in a research lab to improve emergency response during ski races, motivated by a recent cardiac fatality at the American Birkebeiner. Led the software development for a drone-based rescue system, in partnership with DJI, enabling rapid and reliable delivery of medical aid by overcoming the access and response-time limitations of snowmobile-based rescue.
- To make AED drone deployment accessible to non-expert users, designed a simple, user-friendly mobile application integrated with DJI's Mobile SDK (MSDK). Implemented a robust backend path-planning system that supports user-defined no-fly zones and computes optimal flight paths using the A* algorithm.
- Collaborated with the lab director on a technical paper describing the project's methodology, system design, and performance results. Contributed analysis, discussion of impact, and proposed directions for future work.

FTC Team Captain

Sept 2018 – Sept 2025

Techno Maniacs

Acton, MA

- Founded and led my community's only FTC Robotics team for 7+ years, growing it from 4 to 20+ members through mentorship, fundraising, and industry partnerships. Led the team's technical development across autonomous software, localization, computer vision, and custom hardware, ultimately reaching a top-50 global ranking among 7,000+ teams and #1 in New England.

PROJECTS

Open Source Intake | *Java, Android, State Machines, Onshape, 3D Printing, CAM, CNC* **Dec 2024 – May 2025**

- Designed and released a modular, open-source active intake system for the FTC Into the Deep game to lower the barrier to competitive robot design. Optimized roller geometry and flexible mounting enabled fast, reliable pickup without precise alignment across diverse drivetrains.
- Teams adapted and integrated components into their own systems, providing feedback that drove iterative updates and adoption by 100+ teams worldwide, including 3+ at the 2025 World Championship.

SKILLS [STRONG] [STANDARD]

Languages: [Java, Python, HTML/CSS, JavaScript] [Kotlin, C++, C#, TypeScript, Bash]

Concepts: [OOP, Localization, Path Planning, PID] [Sensor Fusion, State Machines, Kinematics]

Frameworks: [React, Android, Unity, Node.js, Next.js] [ROS, Linux, Material-UI, Tailwind, Arduino]

Tools: [Git, Visual Studio, JetBrains IDEs, Onshape, 3D Printing, CNC, Fusion 360] [GCP, Blender, FEA, Isaac Sim]

Libraries: [TensorFlow, OpenCV, DJI MSDK, NumPy] [pandas, PyTorch, face detection, Matplotlib, NLTK]

